DUniversity Clinic Appointment System
Agile Software Solution for Sustainable Development

Course: ITS 610204 / ITS 64604 – Principles of Software Engineering
Semester: February 2026

Sustainable Development Goal: Good Health and Well-Being

Group Members:

Tanzim Daiyan Rahman (0381809) – Group Leader

HU SHUNBO (0391096) – DevOps Engineer

Cui Rongxuan (0391767) – Scrum Master

Sean Yoong Lok Mun (0377520) – Product Owner

Jana Ahmed Mohammed Ahmed Mansour (0391558) – UI/UX Designer

Name	ID	Role	Task
Tanzim Daiyan Rahman	0381809	Group Leader	• ~~Writing test plan~~ • ~~Creating test cases~~ • ~~Collecting user feedback~~ • ~~Recording improvements~~
HU SHUNBO	0391096	DevOps Engineer	• ~~Setting up GitHub repository~~ • ~~Writing README~~ • ~~Organizing folders~~

			• Managing commit history • Compiling report
Cui Rongxuan	0391767	Scrum Master	• ~~Managing Trello Board~~ • ~~Create burndown chart~~ • ~~Monitoring sprint progress~~ • Making sure deadlines are met • Explain Agile methodology used
Sean Yoong Lok Mun	0377520	Product Owner	• Writing Product Vision • ~~Creating User stories~~ • Managing Product backlog • ~~Draw Use Case Diagram~~ • ~~Organizing weekly meeting~~
Jana Ahmed Mohammed Ahmed Mansour	0391558	UI/UX Designer	• ~~Design login page~~ • ~~Design appointment booking UI~~ • ~~Draft class Diagram~~ • Create Wireframe

Abstract

This project is about the University Clinic Appointment System. The system supports SDG 3 – Good Health and Well-Being. It helps improve healthcare services in the university.

The main goal of the system is to reduce long waiting time and make appointment booking easier. Students can register, log in, see available time slots, book appointments, reschedule, or cancel bookings. Clinic staff can manage the schedule and check daily appointments.

We developed this project using Agile and Scrum over four weeks. We created user stories and a Use Case Diagram to define system requirements. We used Trello to manage tasks and GitHub to track our work. We built a working prototype to show the main features. We also tested the system with users and improved it based on their feedback.

This project shows how we applied SDLC phases and Agile methods to build a useful and practical healthcare system.

Project Introduction

Sustainable Development Goal 3 (SDG 3) focuses on Good Health and Well-Being. It aims to ensure healthy lives and improve access to healthcare services for all people. In a university environment, students need fast and reliable access to medical services.

Currently, many university clinics still use manual or semi-manual appointment systems. Students may need to walk in, wait in long queues, or call the clinic to book appointments. This process can cause long waiting times, poor time management, and overcrowding in the clinic.

To solve this problem, our group developed the University Clinic Appointment System. This system provides an online platform for students to book, reschedule, or cancel appointments easily. It also helps clinic staff manage daily schedules more efficiently.

The main objective of this project is to improve appointment management, reduce waiting time, and increase service efficiency in the university clinic. By digitalising the booking process, the system supports SDG 3 and promotes better health management among students.

Agile Process / Trello Usage

This project followed the Agile development approach using Scrum practices. Agile allows the team to develop the system step by step and improve it through continuous feedback.

The project work was organised into short development cycles called sprints. Each sprint focused on completing specific tasks from the Product Backlog. Tasks were prioritised based on importance and system requirements.

To manage the project workflow, the team used Trello as the main project management tool. Trello helped the team organise tasks, track progress, and collaborate more effectively.

The Trello board was organised into different lists, including:

1.Product Backlog – a list of all user stories and system requirements

2.To Do – tasks planned for the current sprint

3.Doing – tasks currently being worked on by team members

4.Done – completed tasks

Each task was represented as a card on the Trello board. Cards included task descriptions, assigned team members, and deadlines. Team members moved cards across the lists as work progressed from To Do → Doing → Done.

This workflow helped the team visualise project progress and identify which tasks were completed or still pending.

Trello also supported Sprint Planning and task prioritisation. The team reviewed the Product Backlog regularly and selected important tasks for each sprint. After completing tasks, the team reviewed the results and made improvements where necessary.

By using Agile practices and Trello, the team was able to maintain clear task management, improve collaboration, and track project progress throughout the development process.

SDLC Phases

1. Planning / Requirements

In this phase, we defined the project goal and aligned it with SDG 3 – Good Health and Well-Being. We have gathered the user requirements and wishes by using Google Form and having

discussions about it. We discover the Functional requirement and non-Functional requirement of the system.

2. Analysis

In this phase, we analysed the system requirements more clearly.

We created a Use Case Diagram using proper UML notation. The diagram includes actors such as Student and Clinic Staff. It also shows the main system functions, including booking appointments, managing schedules, and cancelling appointments.

This helped us understand the functional scope of the system before moving to design.

3. Design

In the design phase, we created wireframes to show how the system interface would look.

We focused on simple navigation and user-friendly layout. The main pages include login, appointment booking page, schedule management page, and confirmation page.

All design files were organised in the GitHub design folder with meaningful commit messages to show progress.

4. Prototype Development

In this phase, we developed a working prototype to demonstrate the core features of the system.

The prototype allows students to log in, select available time slots, and confirm bookings. Clinic staff can view and manage daily appointments.

Development progress was tracked using GitHub commits and linked to Trello tasks.

5. Testing / Evaluation

In this phase, we defined clear test goals, such as checking booking accuracy and system usability.

We conducted basic usability testing with selected users. Users were asked to complete tasks such as booking and cancelling appointments.

Feedback was collected and analysed. Based on the feedback, we improved the interface and fixed minor issues.

This testing process helped ensure that the system is functional and user-friendly.

GitHub Evidence

Prototype Showcase

Prototype Overview

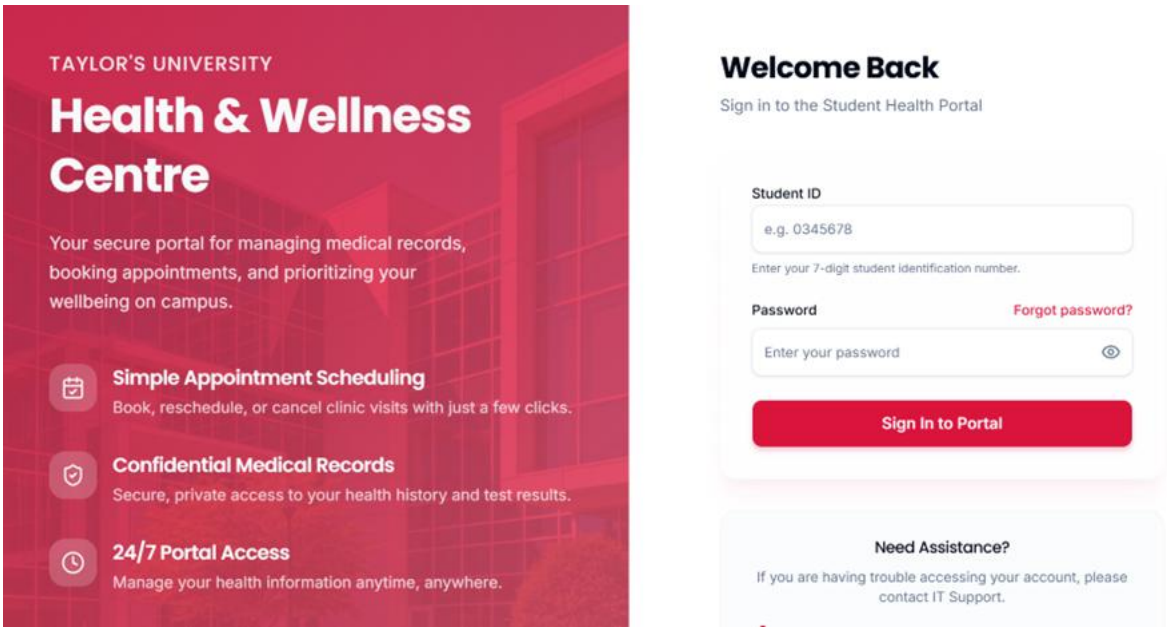
The prototype of the University Clinic Appointment System demonstrates the main features of the system and how students interact with the university healthcare portal.

The prototype focuses on two main functions:
1.Secure student login
2.Appointment booking interface

These features show how students can access the system, schedule appointments, and interact with the clinic services in a simple and efficient way.

The prototype was developed as an early demonstration of the system before full development. It allows users to understand the workflow and evaluate usability of the system.

Login Interface



Screenshot: Login Interface

The User Login Interface Prototype provides a secure way for students to access the university clinic system.

User Login Interface Prototype

The interface is divided into two main sections:

Information Area (Left Side)
This section introduces the university healthcare portal and explains its purpose. It includes:

1. University name and system title

2.Short description of the Health & Wellness Centre portal

3.Key features such as simple appointment scheduling and 24/7 access

4.This section helps users understand the system before logging in.

Login Area (Right Side)
This section allows students to log in using their credentials. It includes:

1.Student ID Field – students enter their university ID

2.Password Field – secure password entry with hidden characters

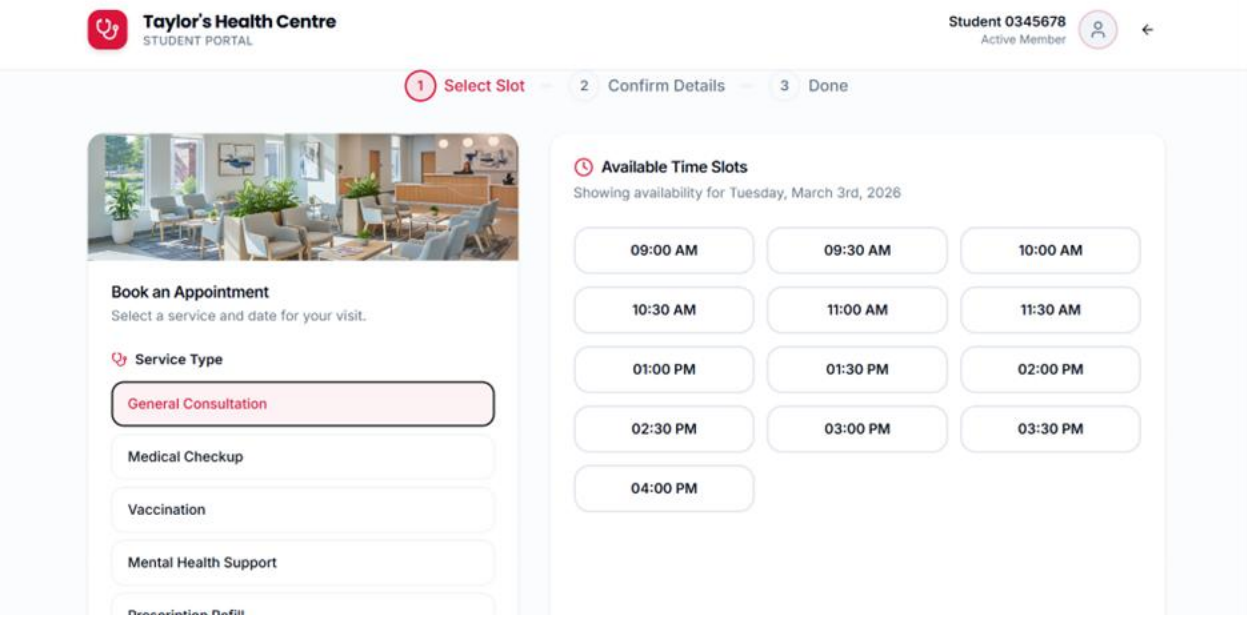
3.Forgot Password Link – allows users to reset their password

4.Sign In Button – submits login information to access the system

5.Support Contact Information – provides help if students face login problems

The login interface ensures that only authorized students can access the system.

Appointment Booking Interface



<u>Screenshot: Appointment Booking Interface</u>

The Appointment Booking Interface allows students to schedule appointments with the university clinic.

The interface provides a simple and clear booking process. Students can:

1.View available appointment time slots

2.Select a preferred date and time

3.Confirm their appointment booking

4.Manage or cancel existing appointments
This interface helps reduce waiting time at the clinic and allows students to plan their medical visits more efficiently.

The design focuses on usability by using a clear layout, readable text, and simple navigation so that students can book appointments quickly without confusion.

Usability Features

The prototype was designed with several usability features to improve the user experience:

1.User Login Interface Prototype

2.Clearly labeled input fields

3.Simple and clean layout

4.Easy-to-read fonts

5.Strong visual contrast for important buttons

6.Minimal interface design to reduce confusion

These features help students interact with the system easily and complete tasks without difficulty.

Security Considerations

Because the system handles student health information, security is important.

The prototype includes several security considerations:

1.User Login Interface Prototype

2.Only registered students can log in

3.Passwords are hidden during entry

4.Unauthorized access is restricted

These features help protect student privacy and medical information.

Reflection & Lessons Learned

During this project, our team learned many important lessons about Software Development Life Cycle (SDLC) and Agile development. Working in a group helped us understand the importance of communication and collaboration. Each member had different responsibilities, and we needed to coordinate our work to complete the project successfully.

Using Trello helped us organise tasks clearly. We created cards for different tasks and moved them from To Do → Doing → Done as the project progressed. This allowed all team members to track progress and understand what tasks were completed or still pending.

GitHub was used to manage our project files and track development progress. By committing updates regularly with clear commit messages, we were able to maintain a clear record of our work and collaboration. This also helped us avoid confusion and manage different versions of files effectively.

Through this project, we also learned how Agile practices such as sprint planning, iterative development, and continuous feedback improve the development process. Instead of building the system all at once, we developed the prototype step by step and improved it based on feedback.

Overall, this project helped us gain practical experience in applying SDLC phases, Agile methods, and software engineering tools in a real development scenario. These skills will be valuable for future software development projects.

Conclusion

This project developed a prototype of the University Clinic Appointment System to support SDG 3 – Good Health and Well-Being. The system aims to improve healthcare access in the university by providing a simple and efficient appointment booking platform.

Throughout the project, our team applied Software Development Life Cycle (SDLC) phases including planning, analysis, design, prototype development, and testing. By following the Agile methodology, the team was able to organise tasks, manage progress, and continuously improve the prototype.

Tools such as Trello and GitHub helped the team manage tasks, track development progress, and collaborate effectively. The prototype demonstrates key features such as secure login and appointment booking, which help reduce waiting time and improve clinic service efficiency.

Overall, this project provided valuable experience in applying software engineering concepts, Agile practices, and teamwork to develop a practical system that supports better healthcare services for university students.

Appendix